// MISSION BRIEFING

ASTROPIXEL MISSION DESIGNER

Build a spacecraft. Fly the whole mission.

See how every part depends on every other.

TEAM SHAY RAFE3 | SPACE MISSION DESIGN GAME | RAMALLAH

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NASA SPACE APPS CHALLENGE 2026

IMPACT

Millions love space. Few see how a mission works

149M
views of NASA's Artemis II coverage, 2026

406.8M
follows of NASA's social accounts, 2024

114,000+
registered for NASA Space Apps 2025

BUT TODAY, ALL THEY CAN DO IS...



WATCH
Livestreams show the launch, not the hundreds of choices behind it



READ
Courses teach one topic at a time, mostly at university



PLAY
Games focus on flying, rarely on budget, power and data

Few tools let you run a whole mission, from first idea to final result.

Sources: NASA Artemis II streaming report, 2026 (views, not people); NASA 2024 year in review (account follows, not unique people); NASA Space Apps 2025 metrics

IMPACT

In Palestine, the gap is even wider



Few aerospace study paths

we found no aerospace engineering degree at a Palestinian university



A trip to NASA for a lucky few

the best-known path takes a small group of teens to NASA Goddard



Mission design is hard to try

astronomy clubs do great work; we add a free mission lab to it

The curiosity is here.

Tools should be too.

CREATIVITY RELEVANCE

AstroPixel: a whole mission, in any browser

1



BUILD: 29 parts based on real hardware, resources live

2



LAUNCH: fly through max-Q

3



CRUISE: real-mission events

4



LAND: Venus, 92 bar, 464 °C

5



LEARN WHY: the cause chain

WHAT PLAYERS DISCOVER +1 camera: +3 kg, +7 W ▸ +5 kg fuel ▸ more data ▸ maybe a bigger battery or rocket ▸ more cost

RELEVANCE VALIDITY

Real physics, NASA data, honest game rules



ROCKET EQUATION

$$\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_0/m_1)$$

Moon: 1,000 kg dry → 1,706 kg fuel

NASA Glenn · Isp 300 s, a GMAT default



SUNLIGHT ∝ 1/d²

Mars 43% · Saturn 1.1%

Weak far out, so RTGs win in-game

NASA Fact Sheets · MMRTG ~110 W at start



SIGNAL

delay = distance / c (one way)

Mars 3–22 min · Saturn ~79 min

NASA Mars Relay Network · MRO 3 m dish



DESIGN MARGINS

35% spare lift · 25% budget

JPL-inspired, our own stated rules

JPL CML checklist (adapted) · Discovery AO

REAL EVENTS 15 stored NASA DONKI storm records (2024), e.g. 10 May, Kp 9. Damage odds are our game rules, and labelled as such.

VALIDITY PRESENTATION

See it fly



SCAN TO PLAY

free · no install · no login

- 8 worlds × 3 tasks
- 1–4 players, one role each

omarshobaki-png.github.io/Shay-Rafe3-Nasa-Space-App-Challenge/play/

Lose your first spacecraft here, for free



Students and space fans

Learn mass, power, data and comms by using them, from zero knowledge

Teachers and schools

One free link for a class: teams of 4 roles, a report that explains every result

Palestinian learners

Any browser on a school PC: no install, no account, no travel

PLAYTESTERS SAID

"It's so detailed."
"It teaches so much about missions and NASA data."

NEXT STEPS

1. Pilot with one school class in Ramallah
2. Act on playtest notes · add NASA photos

HONEST LIMITS

Simplified crew tasks · stored flight paths, no live GMAT · fixed planet distances · English only

TEAM SHAY RAFE3 // THE CREW

Alameen Abu Fkheida
Creative game design & logic

Omar Takhman
Video trailer, design & research

Omar Shobaki
Website, presentation & research

Mohammad Humaidan
Mobile, presentation & research

Mahdi Al Sheikh
Primary game developer

Ahmad Taweel
Website design, game logic & pitch

AI disclosure: the idea and game design are ours; AI helped write code and draft this deck. No AI images, video or audio: all art comes from our own scripts.